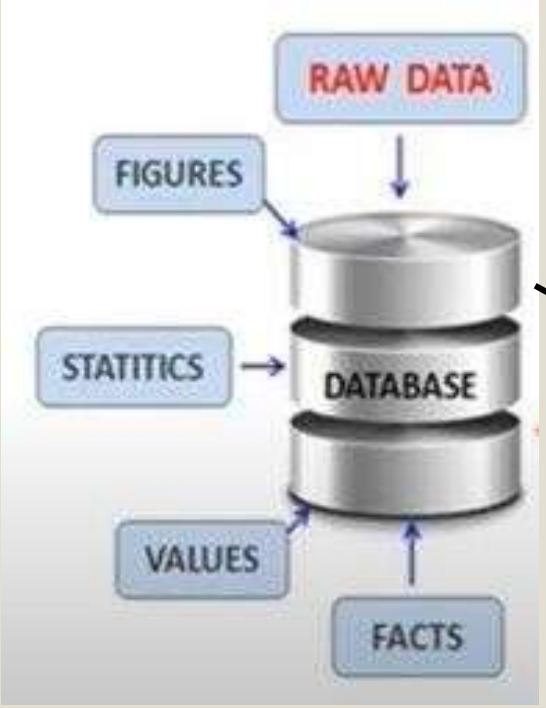
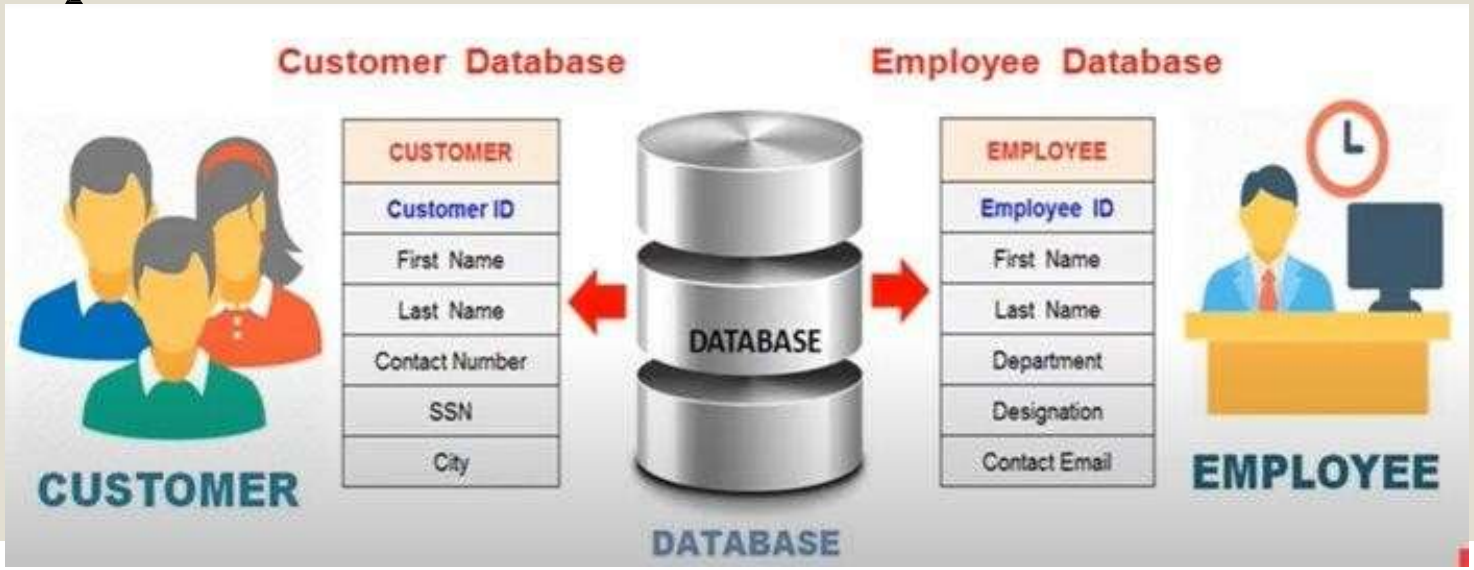
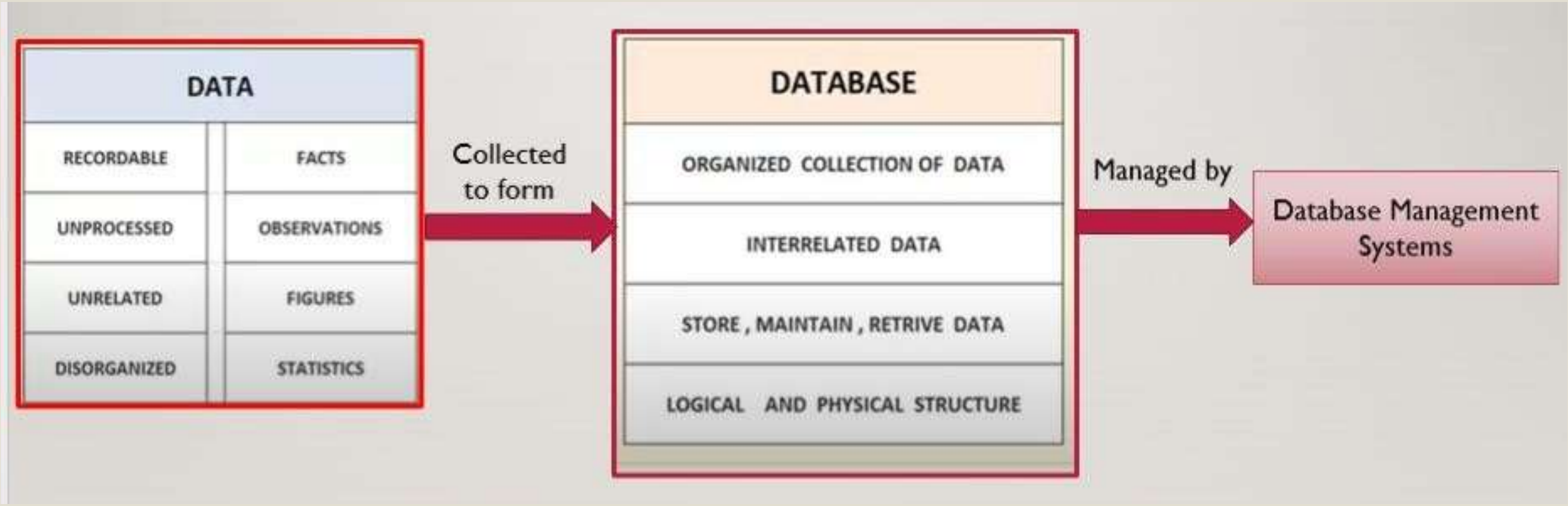
COLLECTION OF DATA FORMS DATABASE





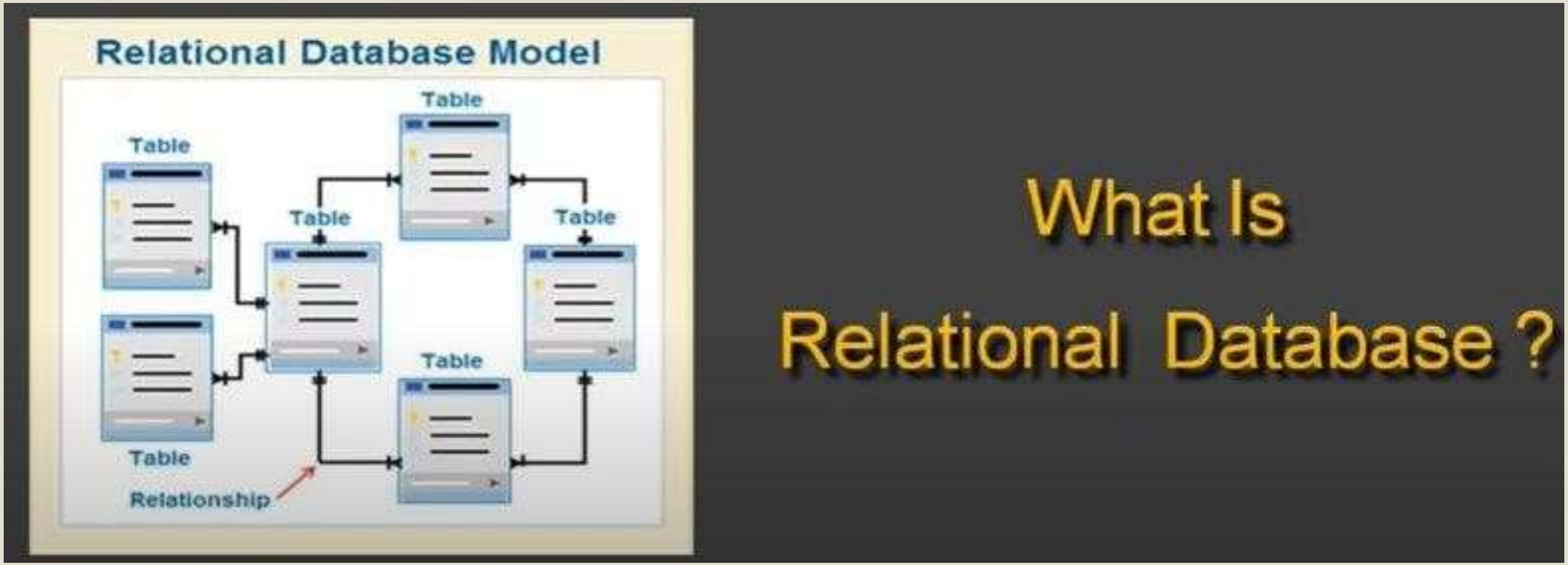
DATA DATABASE DBMS



RELATIONAL DATABASE MANAGEMENT SYSTEM(RDBMS)

- All modern database management systems like SQL, MS SQL Server, IBM DB2, ORACLE, My-SQL and Microsoft Access are based on RDBMS.
- RDBMS is based on relational model introduced by E.F. Codd.
- Software application used to create, manage and administer Relational Databases
- Specially designed for creating and managing relational databases

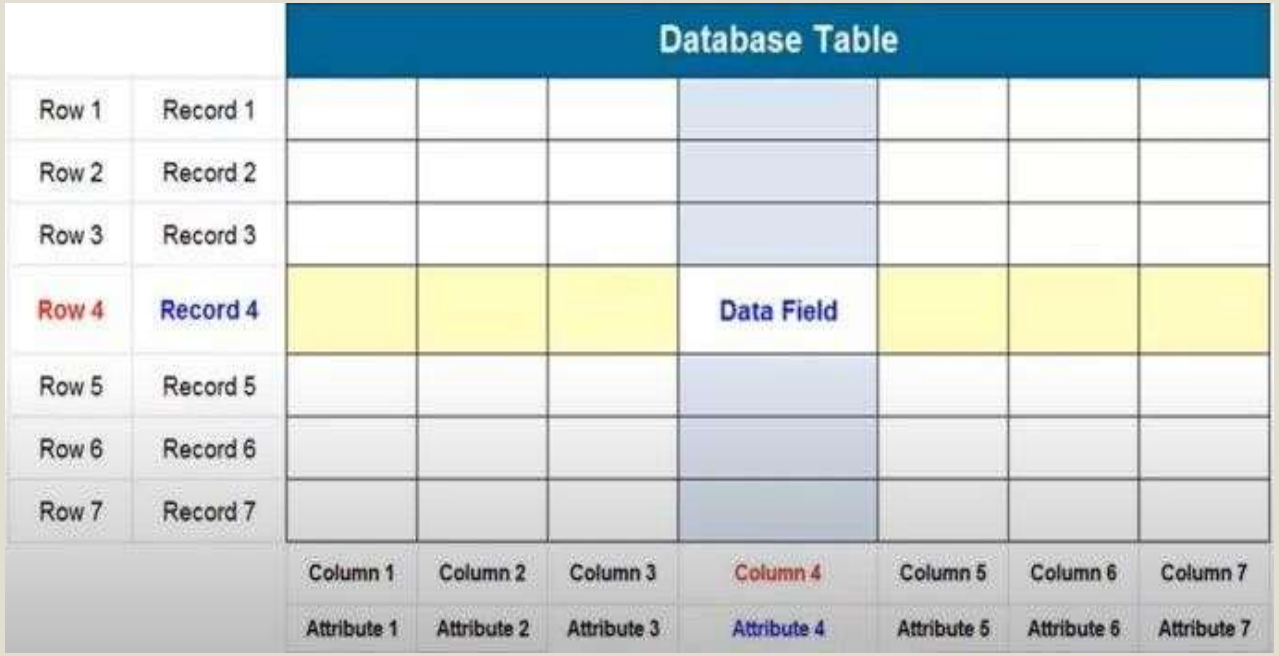
Question?


Relational Database Model
Table
Table
Table
Table
Table
Table
Relationship
What Is
Relational Database ?

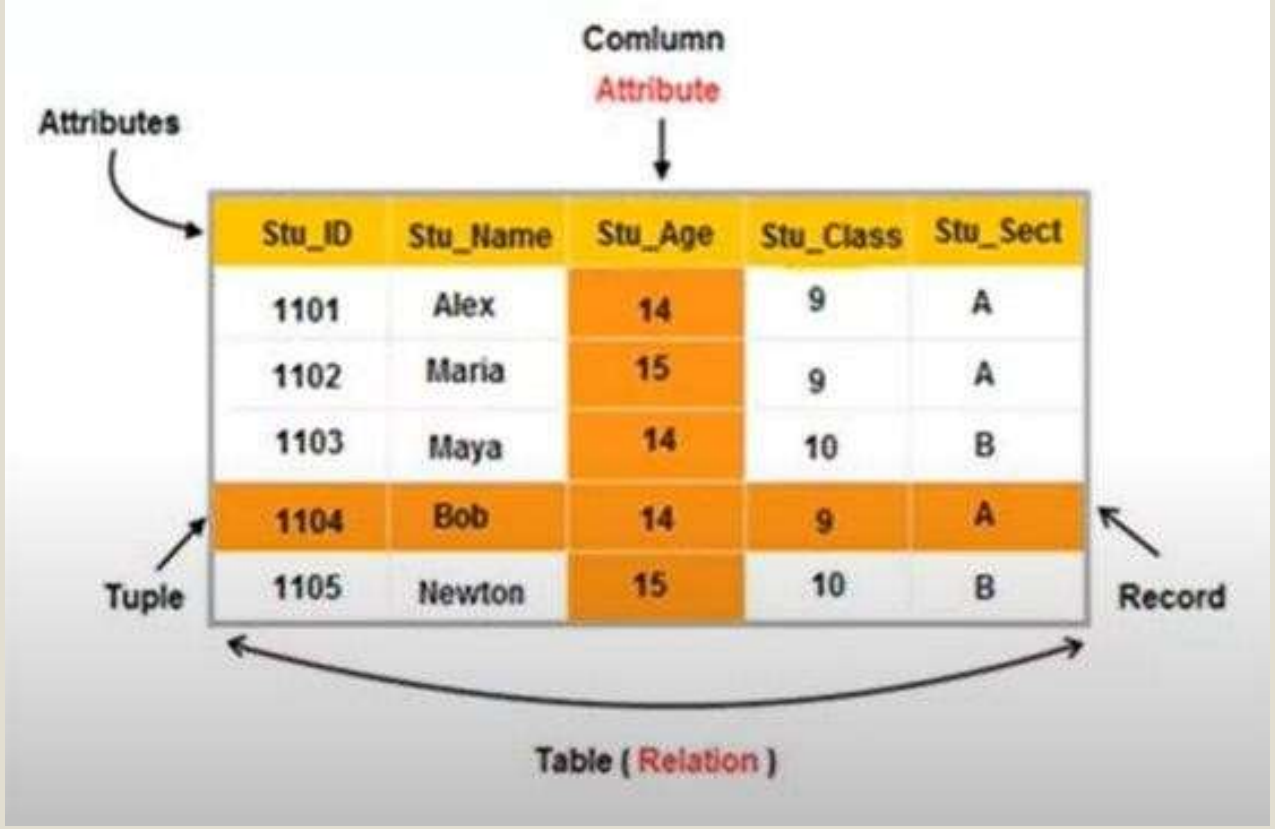
Relational Database

- Database is organized as group of related tables
- Each table represents a database entity

- Each table is a group of column and rows, where
  - column represents attribute of an entity
  - rows represents records.



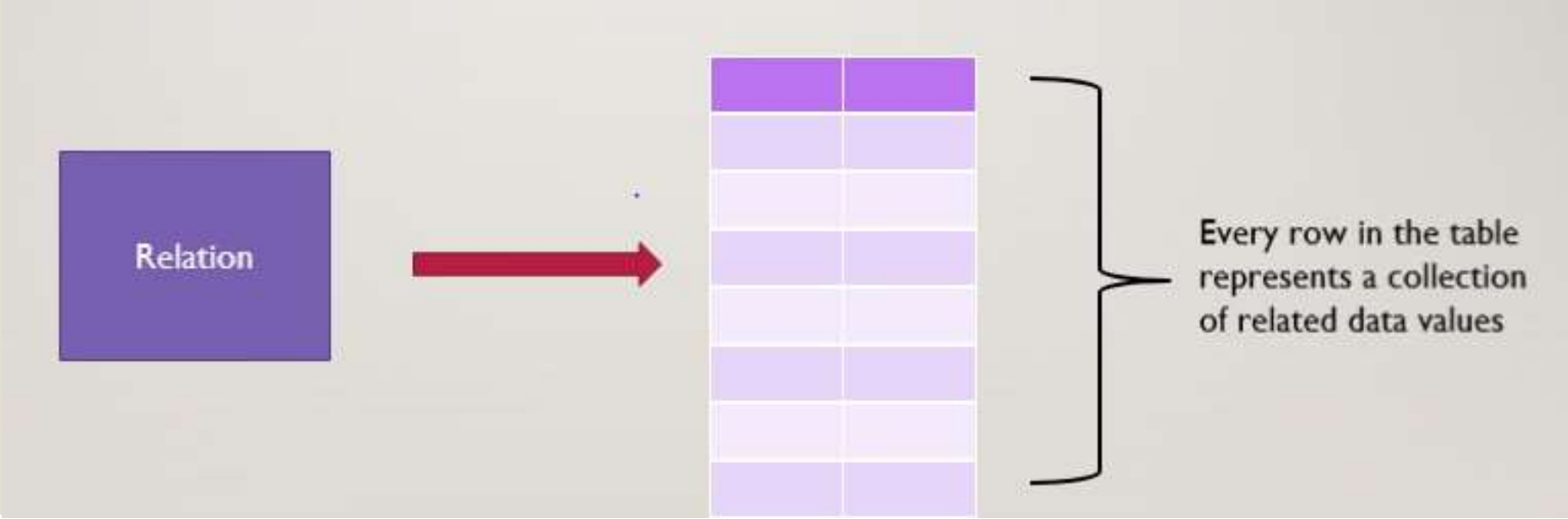
Relational Database - Example



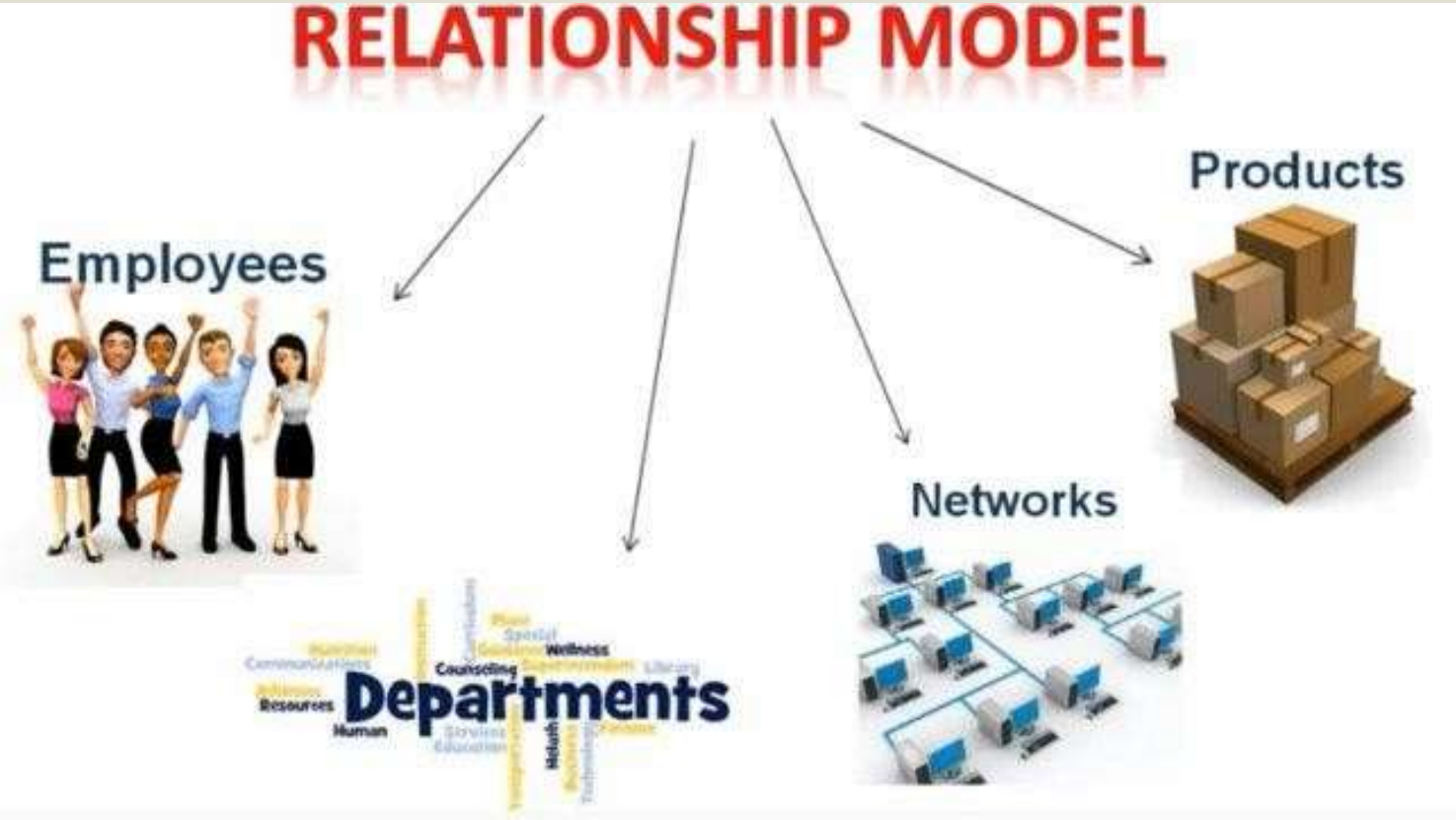
Stu_ID	Stu_Name	Stu_Age	Stu_Class	Stu_Sect
1101	Alex	14	9	A
1102	Maria	15	9	A
1103	Maya	14	10	B
1104	Bob	14	9	A
1105	Newton	15	10	B

Relational Model

- Relational Model represents how data is stored in Relational Databases
- represents the database as a collection of relations
- In relational model, the data and relationships are represented by collection of interrelated tables.



Relational Model - Example



RELATIONAL MODEL TERMINOLOGIES

Table	**collection of data represented in rows and columns**	
Record or Tuple	Each row of a table	
Field or Column name or Attribute	Each column in a Table	

Student_Id	Student_Name	Student_Addr
101	Chaitanya	Dayal Bagh, Agra
102	Ajeet	Delhi

Student_Id	Student_Name	Student_Addr
101	Chaitanya	Dayal Bagh, Agra
102	Ajeet	Delhi

Student_Id	Student_Name	Student_Addr
101	Chaitanya	Dayal Bagh, Agra
102	Ajeet	Delhi

RELATIONAL MODEL TERMINOLOGIES

Domain	set of permitted values for an attribute in table	For e.g., domain of month-of-year can accept January, February,...December
Degree	total number of attributes	-
Cardinality	total number of rows present in the Table	-
Relation Schema	represents name of the relation with its attributes	e.g.; STUDENT (ROLL_NO, NAME, ADDRESS, PHONE and AGE) is relation schema for STUDENT
NULL Values	value which is not known or unavailable is called NULL value	represented by blank space

RELATIONAL MODEL TERMINOLOGIES

relation instance	**specific instance of a relation, i.e., containing a specific set of rows**	
Keys	they can uniquely identify the tuple	

RELATION SCHEMA

- Database Schema
  - logical design of the database
- Relation Schema
  - consists of a list of attributes and their corresponding domains

◦ A1, A2, …, An are attributes

◦ R = (A1, A2, …, An) is a relation schema

◦ Example:

◦ instructor = (ID, name, dept_name, salary)

◦ A relation instance r defined over schema R is denoted by r (R).

◦ The current values a relation are specified by a table

RELATION SCHEMA

dept_name	*building*	*budget*
Biology	Watson	90000
Comp. Sci.	Taylor	100000
Elec. Eng.	Taylor	85000
Finance	Painter	120000
History	Painter	50000
Music	Packard	80000
Physics	Watson	70000

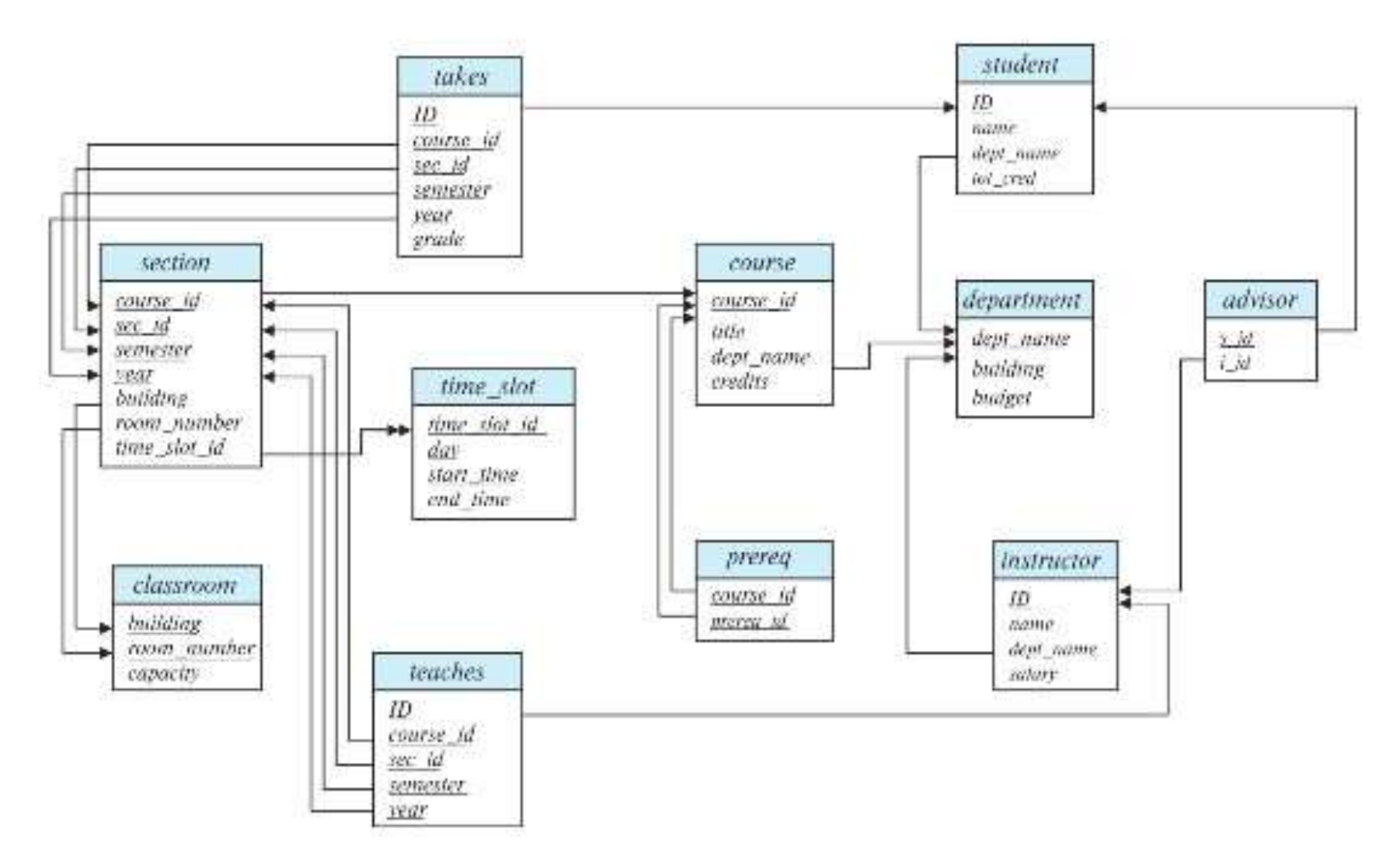
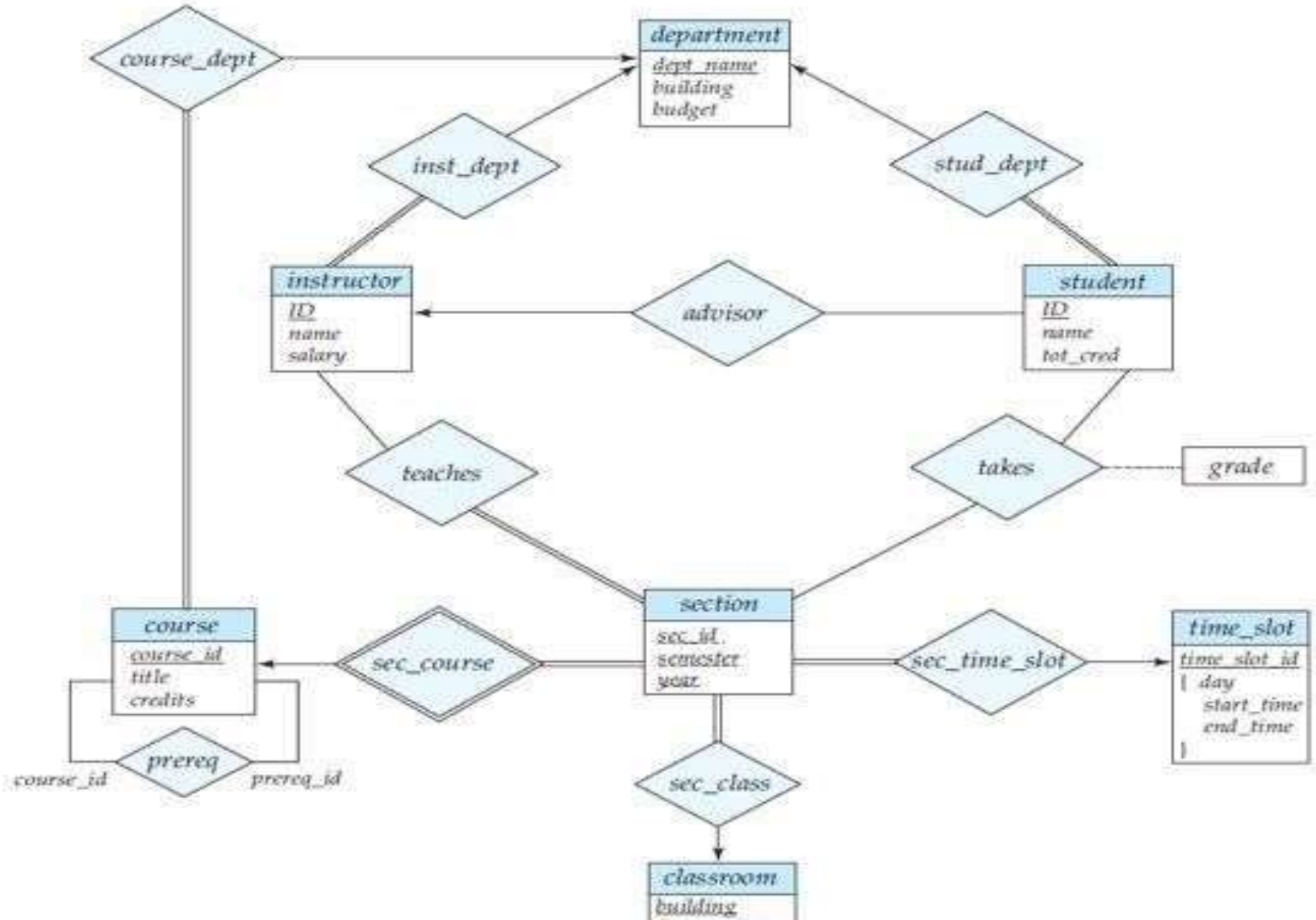
The *department* relation

department (*dept_name*, *building*, *budget*)

Schema Diagram

- A database schema, along with primary key and foreign key dependencies, can be depicted by schema diagrams
- Each relation appears as a box, with the relation name at the top, and the attributes listed inside the box
- Primary key attributes are shown underlined
- Foreign key dependencies appear as arrows from the foreign key attributes of the referencing relation to the primary key of the referenced relation.

Schema Diagram for University DB



Write the relational schema?